DEPARTMENT OF COMPUTER ENGINEERING
BE/Sem-VIII/Jan-Apr 2026 Course Code: CSL8023
Subject : APPLIED DATA SCIENCE LAB
Experiment List

Select a case study and formulate the problem statement for a real world Application . Implement experiments listed below on the selected case study. [*Suggested Case Studies: Customer Segmentation, Fraud Detection, House Price prediction, Product Recommendation, Stock price prediction , Weather prediction etc.,***]**

No	Title	CO
1	Explore the descriptive statistics on the given dataset.	CO1, CO2
2	Apply data cleaning techniques (e.g. Data Imputation).	CO1,CO2
3	Explore data visualization techniques.	CO1,CO2
4	Implement and explore performance evaluation metrics for Data Models (Supervised)	CO1,CO3
5	Implement and explore performance evaluation metrics for Data Models (Unsupervised Learning)	CO1,CO3
6	Implement time series forecasting.	CO1,CO3
7	Outlier detection using distance based/density based method.	CO1,CO3
8	Use SMOTE technique to generate synthetic data.(to solve the problem of class imbalance)	CO1,CO2
9	Inferential statistics -Hypothesis Testing, Z test, T-Test , Anova	CO1, CO2
10	Demonstration of Selected Case study	CO1,CO2, CO3
	Assignments	
1	Assignment on visualizing high dimensional data:-Parallel chart, Deviation chart, Andrews Curves.	CO1,CO2
2	Assignment on data science analysis and visualization tools-SPSS,Tableau, Knime,Google Analytics, BigML…etc	CO1, CO3

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 01

AIM: Explore the descriptive and inferential statistics on the given dataset.
THEORY:

1. Descriptive statistics

a) measures of central tendency -mean, median,mode, min,max, sum

b)measures of dispersion -range, IQR, standard deviation, variance, Skewness correlation

Visualize your data using Box-and-Whisker Plot,Scatter Plot, Correlation Matrix

2. Inferential statistics -

a)Distributions -Normal,poisson, Population Parameters, Sampling Errors Confidence Intervals

b)principles of hypothesis testing -Z-test , t-test,type I and Type II error, Anova

Hypothesis testing:

		Decision	
		H_0 true (Fail to reject)	H_0 false (Rejecting H_0)
Actual	H_0 true	TRUE NEGATIVE Correct decision: **Confidence level** (prob $1-\alpha$)	FALSE POSITIVE Type I Error: **Significance level/Size (α)** (prob α)
	H_0 false	FALSE NEGATIVE Type II Error: fail to reject (prob β)	TRUE POSITIVE Correct decision: **Power** (prob $1-\beta$)

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 04

AIM- Implement and explore performance evaluation metrics for Data Models (Supervised)

Theory -
Classification -
1)Evaluate models using the following metrics

Metric	Formula	Evaluation Focus
Accuracy	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$	Overall effectiveness of a classifier
Error rate	$ERR = \frac{FP+FN}{TP+TN+FP+FN}$	Classification error
Precision	$PRC = \frac{TP}{TP+FP}$	Class agreement of the data labels with the positive labels given by the classifier
Sensitivity Specificity	$SNS = \frac{TP}{TP+FN}$ $SPC = \frac{TN}{TN+FP}$	Effectiveness of a classifier to identify positive labels How effectively a classifier identifies negative labels
ROC	$ROC = \frac{\sqrt{SNS^2+SPC^2}}{\sqrt{2}}$	Combined metric based on the Receiver Operating Characteristic (ROC) space [53]
F_1 score	$F_1 = 2\frac{PRC \cdot SNS}{PRC+SNS}$	Combination of precision (PRC) and sensitivity (SNS) in a single metric
Geometric Mean	$GM = \sqrt{SNS \cdot SPC}$	Combination of sensitivity (SNS) and specificity (SPC) in a single metric

Recall=sensitivity

False positive rate (α) = type I error = 1 − specificity = FP / (FP + TN)

False negative rate (β) = type II error = 1 − sensitivity = FN / (TP + FN)

Power = sensitivity = 1 − β

2) Plot ROC curve

Regression

1)Evaluate models using the following metrics

Evaluation Metrics	Formula	Layman's explanation
Pearson correlation coefficient (r)	$r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2}\sqrt{(Y-\bar{Y})^2}}$ Where, $\bar{X}$ - mean of X variable $\bar{Y}$ - mean of Y variable	Measure of the strength of the association between the two variables. In terms of predictive accuracy of the model, i.e., the Pearson correlation coefficient (r) between the predicted values and the observed values.
Coefficient of determination (R2)	$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	Equals the fraction of variance of outcome/response explained by the model. Also, mostly referred to as goodness of fit.
Mean Squared Error (MSE)	$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	Most commonly used loss function for regression. It tells you how close a regression line is to a set of data points.
Root Mean Squared Error (RMSE)	$\text{RMSE}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	Estimate to measure the error of the numerical predictive model. In short, measure the error rate of a regression model. Also, commonly referred as Root Mean Square Deviation (RMSD).
Root Mean Squared Relative Error (RMSRE)	$\text{RMSRE} = \sqrt{\frac{\sum_{i=1}^{s}\left(\frac{y_i-\hat{y}_i}{y_i}\right)^2}{s}}$	Here the error will be considered relatively which means that each residual is scaled against actual value or normalized by that. A residual is the vertical distance between a data point and the regression line. Each data point has one residual. Lower the value better we can establish the quality of model prediction.
Mean Absolute Error (MAE)	$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	It is very similar to MSE in the sense that here also we take the difference between the predicted and the actual value, and divide it with the number of testing sample values, however, instead of squaring this difference, we take an absolute value of it.
Mean Absolute Percentage Error (MAPE)	$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$	Also, used as a loss function for regression problems. It measures the accuracy as a percentage. As MAPE is the measure of error, higher values denote higher error, while lower values represent good accuracy. Also, commonly referred as Mean Absolute Percentage Deviation (MAPD).

where y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean value of the observed values, **s/n** denotes the sample size of the testing dataset.

Decision analysis

SWOT analysis
decision tree analysis
cost-benefit analysis
expert analysis

Class: BE-Computer; Semester: VIII
Subject: Applied Data Science Lab
Experiment No.: 05

AIM- Implement and explore performance evaluation metrics for Data Models (Unsupervised)

Theory -
Unsupervised learning algorithms

- **K-means clustering**
- **KNN (k-nearest neighbors)**
- **Hierarchal clustering**
- **Neural Networks**
- **Principle Component Analysis**
- **Apriori algorithm**

Clustering - 1)Evaluate models using the following metrics

Adjusted Rand index

$$\mathbf{E}[\mathrm{RI}] = \frac{\text{Number of pairs in same cluster (actual)} \times \text{Number of pairs in same cluster (predicted)}}{\text{Total number of possible pairs}}$$

$$= \frac{\sum_i \binom{a_i}{2} \times \sum_j \binom{b_j}{2}}{\binom{n}{2}}$$

$$\text{Adjusted Rand Index (RI)} = \frac{\text{Number of pair-wise true positive prediction} - \mathbf{E}[\mathrm{RI}]}{\text{Average number of pairs in same cluster for actual and predicted} - \mathbf{E}[\mathrm{RI}]}$$

$$= \frac{\sum_{ij} \binom{n_{ij}}{2} - \mathbf{E}[\mathrm{RI}]}{\frac{1}{2}\sum_i \binom{a_i}{2} \times \sum_j \binom{b_j}{2} - \mathbf{E}[\mathrm{RI}]}$$

Silhouette Coefficient

$$\mathrm{a} = \text{Average distance between sample and all other points in same cluster}$$

$$\mathrm{b} = \text{Average distance between sample and all other points in next nearest cluster}$$

$$\text{Silhouette Coefficient } (s) = \frac{b - a}{\max(a, b)}$$

2) For clustering - plot sum of squared distance(sse) against k , use Elbow method

https://scikit-learn.org/stable/modules/clustering.html#clustering-performance-evaluation

Mutual Information based scores

Homogeneity, completeness and V-measure

Association